

Designing and Implementing Individualized VR for Supporting Depression

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ABSTRACT

Mental health conditions are one of the main struggles faced by many in the society, posing a challenge on the healthcare system. In New Zealand alone, about one in five people will suffer a serious mood disorder such as depression at some point in their life. Studies suggest that the use of VR (Virtual Reality) technology has the potential to be an effective tool in the treatment of mental health conditions like anxiety, eating disorders, and phobia. However, the subject of the therapeutic use of VR to treat and prevent depression has not been widely studied. Other than that, there is also little known about how individualizations in VR can affect user experience to improve mental health. In our previous work, we have presented the design and architecture of iVR, an individualised VR experience to enhance self-compassion, a protective factor in the treatment and prevention of depression. In this paper, we outline the process of implementing the design and the result of our recent feasibility study. The majority of the participants agreed that introducing elements of individualization within iVR positively impact their user experience and that the iVR can potentially enhance their self-compassion. We reached out to seven mental health professionals for their feedback, and most felt that introducing elements of choice in the VR experience would help them increase their knowledge of clients. We believe that this project can pave the way for larger-scaled efficacy testing for clinical use of intelligent individualized VR technology in the treatment and prevention of mental health conditions in the future.

Keywords: Virtual reality, individualisation, mental health, user models, user experience, feasibility study, self-compassion, depression.

1 INTRODUCTION

Mental health disorder is one of the major challenges faced by many in the society, and one of the major burdens in the healthcare system globally [1]. According to the World Health Organization [1], conditions related to mental, neurological, and substance disorders account for 14% of the global burden of disease, a number

predicted to rise significantly by the year of 2030, putting a strain on the healthcare system in countries that may not have the capacity to reach the majority of those who needed care.

In New Zealand alone, nearly a quarter of the population have poor mental wellbeing [2] and one in seven people will have experienced a major depression by the age of 24 [3]. As mental health disorders can potentially lead to lifelong consequences, early intervention can positively improve the prognosis in this case [4]. Therefore, it is increasingly important that care is available and easily accessible for those who needed it.

One method used to improve symptoms of depression is Compassion Focused Therapy (CFT). The link between self-compassion/self-criticism to depressive symptoms have been studied extensively [5], with self-compassion and self-criticism being protective and risk factors respectively. Positive self-compassion was described as treating oneself with kindness and understanding during times of struggle [6], as opposed to self-criticism, which was characterized by negative self-judgement and self-evaluation in perceived failure [7]. Higher level of self-criticism and lower level of self-compassion are linked with higher depressive symptoms and have been found to increase one's risk of chronic or repeated depression over the course of their lives [8].

A study by Jazaeri, et al. [9] suggests that compassion can be increased via training. The aim of CFT is to increase the client's compassion through training by strategically exposing them to situations where a degree of compassion towards themselves and/or others is required and encouraging the client to focus on understanding and feeling self-compassion in negative thought processes [10]. However, though it is effective, simulating such scenarios can prove to be challenging and resource demanding, therefore there is a need for innovative interventions that are more sustainable and easily accessible [11].

Virtual Reality is an emerging and innovative technology that can provide immersive and realistic simulated experiences in a virtual world. Because of its growing popularity, VR Head Mounted Displays (HMD) is becoming more common, making it more affordable and accessible for the masses. Recently, there is a growing interest in using Virtual Reality as a therapeutic tool for mental health conditions. In virtual reality simulations, scenarios and experiences can be recreated artificially, providing realistic simulations of real-life scenarios in a low-risk environment, allowing explorations of different coping strategies and symptom management techniques for people with mental health issues without extensive and costly external resources.

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Recent studies support the efficacy of VR in the treatment of mental health disorders, addiction, pain management, and eating disorders where progress made from the virtual exposure therapy seems to apply in real-life situation [12]. In the treatment of addiction, VR has been used to simulate a situation where addiction behaviour might occur so that therapist can observe the client's behaviour and reaction when exposed to the situation and use the information to plan an appropriate therapy for the patient [13]. VR has also had positive results in the treatment of phobia, in particular the treatment of acrophobia, panic disorder, trypanophobia, agoraphobia, claustrophobia, agoraphobia, social phobia, fear of driving and arachnophobia [14]. VR can be used to provide exposure therapy in a safe and controlled manner in the treatment of phobia [15]. VR has also been used as exposure therapy in the treatment of patients with post-traumatic stress disorder (PTSD), where patients are exposed to the situations that triggers their trauma in combination with relaxation training to help them adapt to the stress, decreasing their anxiety when exposed to a triggering situation over time [16]. VR technology was also used to facilitate cognitive rehabilitation in individuals with autism to help them gain more independence by improving their essential skills such as picking up visual cues, paying attention to other parties in conversations, and crossing a road [13]. However, despite the numerous studies of the use of VR to support mental health, there is little known about its efficacy in the treatment of individuals with depression [17]. There is also very little work done on investigating the possible benefits of the individualization of virtual experiences for mental health therapy, for instance, tailoring the virtual environments based on the subject's personal interests and preferences. In this project, we seek to investigate if introducing an element of choice and individualization can provide benefits to individuals with mental health issues as well as to their therapists.

2 PILOT STUDY

A pilot study was recently conducted with the sample size of 23 (5 females and 18 males, mean age of 24) [19]. The study was done with an earlier iVR prototype, with the goal to gather data on whether participants enjoyed interacting with the app, whether the introduction of choices improves user experience, and if it can be potentially useful for increasing people's self-compassion and improve their mental wellbeing in the long run [19]. Participants were asked to fill out a Patient Health Questionnaire 9 (PHQ-9), a measure often used to assess and monitor depressive symptoms and the outcomes in psychological treatments for depression [20]. The participants then interacted with the iVR prototype and filled out a subjective questionnaire at the end. Findings from our pilot study show that regardless of their PHQ-9 scores, most participants feel that the introduction of individualization features makes the iVR app more engaging and that it has the potential to be used in the real world [19]. We also reached out to seven clinical psychologists to gather their feedback on the iVR application [19]. Most of the mental health professionals approached believed that the introduction of individualization on the iVR app would help them increase their knowledge of clients and that it has the potential to be used in a clinical setting.

3 iVR DESIGN AND IMPLEMENTATION

In our initial study [18], we focused on designing and implementing a VR application to aid recovery from depression by improving self-compassion, expanding further on findings in an earlier study by Falconer et al. [17] on whether social interaction in an immersive virtual reality could improve self-compassion in patients with depression. In Falconer's study [17], participants are presented with a figure of a crying child in a virtual environment where they were encouraged to demonstrate compassion towards the child, and in turn, increase their own self-compassion. The study shows a

substantial increase in participant's self-compassion after exposure to VR mental health therapy.

In our previous work [18], elements of individualization were introduced in the design of iVR, a prototype of individualized virtual reality for mental health. Participants were allowed to choose their own avatar, therapy environment, and scenario. Therapists could observe the individual's choices and use them as discussion points, thus supporting therapist in the process of gaining knowledge on their patients, who might not be as forthcoming in giving information related to their mental health or struggle to elaborate their life situation and underlying issues.

Findings of our pilot study [19] suggested that individualization of Virtual Reality mental health therapy can be beneficial for patients and therapists by increasing self-compassion in patients, enhancing user experience, and helping therapists to gather knowledge on their patients that would otherwise be difficult to obtain. However, there were some aspects of the prototype that can be improved on based on participants' feedback, mainly in terms of the graphics and UX/UI [19].

The requirements and sequences of scenarios in this iVR application follows the specifications outlined in our earlier proof of concept study [18], considering the feedback gathered from participants [19]. Participants will be presented with three different views in total.

In one session, participants will first be presented with a main menu stage that enables them to customize their virtual body, in-game character, environment, and scenario with a GUI panel.

Available choices of virtual bodies are as follows:

- Asian male
- Asian female
- Pakeha male
- Pakeha female
- Maori male
- Maori female

Choices of therapy environment are as follows:

- Living room
- A study room in a Victorian house
- Park

Available in-game characters are as follows:

- Young female
- Young male

Available scenarios are as follows:

- Delivering compassion to a crying person
- Delivering compassion to an angry/upset person

Interactions with the main menu will be done via Oculus Quest 2 hand-held controllers. When they are done selecting all available choices, they can then proceed to the first scene where they can practice delivering self-compassion to a virtual character.

Like the original virtual experience in Falconer's study [17], they will be delivering compassion to an in-game character from a virtual body on the first scene. However, unlike the pre-determined scene in Falconer's study [7], the scene will be generated based on the participant's choices from the main menu. Instead of a pre-determined virtual body, the appearance of avatar will be determined based on the chosen virtual body. The therapy environment, in-game character's appearance and behavior will follow participant's choice as well. When the scene finished loading participant's choices, they can start speaking to deliver compassion. The speech of the participants in this scene will be recorded to be used on the second scene. When they have finished delivering compassion, they can go to stage by clicking the "Next Stage" button.

The following figure is a screenshot from scene 1 when a participant chose to engage with an upset young female with a female avatar of Asian ethnicity in the living room environment.



Figure 1: Participant giving compassion on scene 1.

On the second stage, the participants will experience receiving compassion from their chosen virtual body with the point of view of the in-game character (as shown in Figure 2). The scenario ends after the avatar finished reenacting the scene and they can choose to go to the main menu or quit the application.



Figure 2: Participant's point of view on the second scene.

3.1 User Interface

When designing the main menu for the virtual environment, there were some things that we took into consideration. The first point of consideration is the comfort of the end user. Things that work well for the UI of a traditional 2D surface might not necessarily be applicable for the UI of a virtual environment [21]. For example, a carousel or scrollable menu might work well for a traditional 2D surface and are intuitive. However, this can present some discomfort if not implemented correctly in a 3D virtual environment.

The second point of consideration is how easily can end users see the menu without having to do unnecessary navigation. In a virtual world, users are free to interact and view the environment from any angles. The menu should be clearly visible within their field of view. Some UI formats might be more difficult to navigate than others. For example, in a virtual world, users might find it difficult to see the full extent of the menu when they are presented in a 3D carousel format. They might need to walk around to find the items or options to select or interact with, and this is not always practical. In this case, scrollable menu or panel might be a better option. With scrollable menu, users will be able to see the whole menu options easily in one place without having to spend extra effort to look for it. However, a smooth scrollable menu might cause discomfort as the scrolling motions can cause users to feel like there were unintentionally moving in the environment as they navigate through the menu [21]. To minimize discomfort, we reduced the amount of movement in the menu by presenting available options on a paginated panel within the camera frustum and toggles the pages of the panel depending on the player's input. Figure 3 shows some choices of therapy environments displayed on a panel.

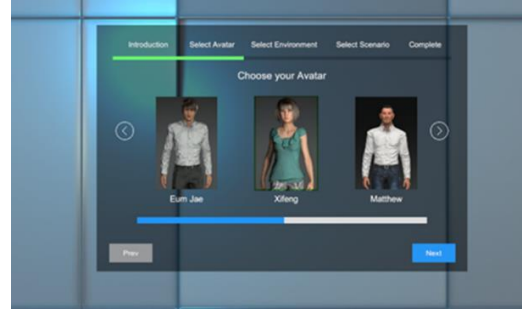


Figure 3: Avatar selection on the Main Menu panel.

3.2 Virtual Environments

In the older prototype version of iVR [18], some elements of individualization were introduced by allowing participants to choose from a variety of therapy environment. However, these environments were in their early prototype phase and does not provide a visually realistic representation of what the environment should look like (as shown in Figure 4). At the end of the pilot study, it was inferred from the qualitative data analysis of the participants' questionnaire feedback that participants tend to favour environments that looks more realistic as it can enhance their sense of immersion in the therapy environment [19].

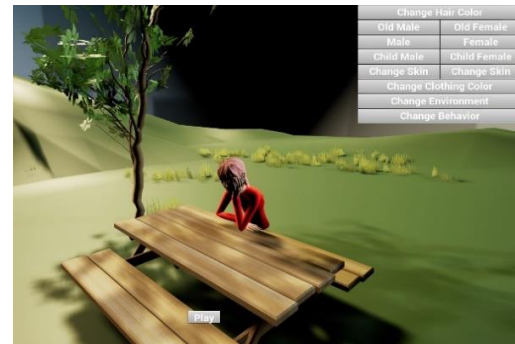


Figure 4: Environments from Baghaei's initial iVR prototype.

At this phase of the project, we focused on improving the graphical quality of the selectable environments to better reflect real life environments and make it more immersive for participants. When developing the virtual environment, we need to take into consideration the balance between performance and complexity of the assets. High fidelity assets can provide a visually more realistic environment. However, it can be detrimental to the app's performance as it increases the burden on the software and hardware of the VR HMD to load the digital models. To prevent performance issues, we incorporated 3D digital models that are reasonably photo realistic yet are not too complex as to hinder the performance of the software. We also used occlusion culling to optimize performance by not rendering things that are out of view.

From our pilot study [19], it was found that participants liked the inclusion of indoor and outdoor sceneries and the ability to change sceneries. Based on this feedback, we included both indoor and outdoor environments for the participants to choose from in the recent version of the iVR Prototype: 1) Living room of a house, 2) A study room, 3) Park. Figure 5-6 shows the improved assets in the outdoor and indoor environments from our iVR prototype.

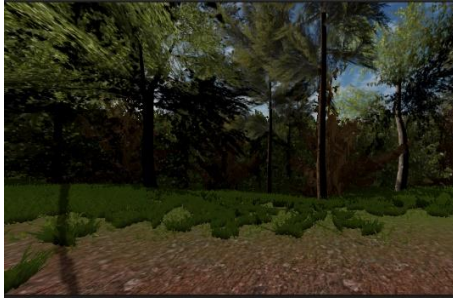


Figure 5: Improved outdoor experience in the iVR prototype.



Figure 6: Indoor assets used in the iVR prototype.

3.3 Personalized Avatars

In the earlier prototype [18], it was found that participants much preferred more graphically realistic avatars to interact with. Therefore, at this stage of the project, we aim to improve the avatar's look and feel. Creating a convincingly realistic humanoid model can be challenging as humans are graphically complex objects, and the human eye is very good at detecting subtle nuances of humanoid objects [22]. Modelling humanoid models can be broadly categorized into two, the surface model that consists of the skin layer and skeleton layer and the multi-layered model that consists of the skeleton layer, an intermediate layer muscle tissues and bones that simulate movements, and a skin layer [22]. An example of skeleton rigging on the multi-layered model layer is shown in figure 10.



Figure 7: Skeleton rigging on humanoid models.

For the scope of this project, we used pre-existing multi-layered humanoid assets created on the Realusson software. These assets are rigged with a bone structure so that it is easier to animate the behaviours of the avatar.

3.3.1 Animation

One feedback from our earlier prototype was that the avatar movements does not feel very natural, and that the participants wanted to interact with avatars with a more human-like behaviour.

According to O'Brien and Hodgins [23], there are three basic categories of techniques that can be used to animate human behaviours: 1) Keyframing, 2) Motion capture, and 3) Simulation. These techniques each has some advantages and disadvantages in terms of level of control over motion details, cost of production, performance, and reusability [24]. Keyframing technique requires manually specifying critical or key positions of objects in each timeline. With this technique, animators can have more control over the subtle details of the object's motion. It also allows for a smoother transition between poses. However, the downside is that it can be time consuming. In motion capture, the real object's movements are captured and recorded in a computer-usable form. This might be less time consuming than the keyframe method, but it does not give much control over the motion and does not necessarily have smooth transitions between scenarios, so in our case, this might not be ideal. The third technique is simulation, where motion is generated using the laws of physics. While this method can generate realistic movements, it might be very costly and time consuming, thus is not feasible at this stage of the project. After weighing in the pros and cons of each technique, we decided to use the keyframing technique to allow for more control and smoother transitions between scenarios.

We used Mixamo and Blender software to animate our avatars. The models were rigged with a bone structure. When the animation process is done on Mixamo, it is edited on Blender software if it needs some finer animations (as shown in Figure 8), then exported to an FBX file so that it is ready to use in the Unity project. Afterwards, we defined the animation transitions and pose orders via animation controllers in Unity. At this stage, we provided two behaviours for participants to choose from: crying and standing around upset.



Figure 8: Keyframing animation process in Blender.

3.3.2 Facial Expression and Audio Recording

On each session, iVR records the Oculus headset's own mic input from scene 1 and uses the recording as the audio input for scene 2 so that the end user can hear and see their avatar re-enacting scene 1 from another point of view. To control the avatar's facial expressions, we used OVRipSync, a Unity add-on plugin developed by Oculus that can be used to sync lip and facial movement of a digital asset to match audio input.

4 CONCLUSION

This paper outlines the process of designing and implementing an enhanced prototype of individualized virtual reality application to support depression.

The next phase of this project would be to gather data from a bigger sample size on the impact of adding elements of individualizations on the self-compassion in patients with

depression. To verify if the addition of elements of individualization positively impacts self-compassion, we aim to recruit 60 participants aged 18-24. Participants will be randomly divided into a control group and experimental group. The control group will be using iVR without elements of individualization, while the experimental group will be using iVR with individualization.

The method of data gathering will be partially based on Falconer's original feasibility study [17]. Self-compassion will be measured using Self-Compassion and Self-Criticism Scale (SCCS) [11] and depression will be measured using the Patient Health Questionnaire-9 (PHQ-9) [8]. Each group will have six sessions over three weeks. We will record their depression and self-compassion once before the test and once more after the test. We will then compare and analyse the measurements pre-test and post-test to see if there are a significant improvement in the measurements of depression and self-compassion.

Going forward, we are considering some possible enhancements to the iVR application by introducing more robust interaction between participants and the virtual environment, focusing mainly on the interactions with the avatars present in the environment. We are looking into adding some interactive objects that can be used by participants to encourage user engagement. For instance, participants can pick up virtual objects that can provide comfort such as a tissue box or hot chocolate via controller hands and offer them to the avatar. The avatar will then react either positively or negatively based on the scenario. Another thing that can be done to improve user engagement is adding some artificial intelligence to the avatar. This allows more interactions with the avatar. Our hypothesis is that introducing interactive elements into the application will increase the efficacy of iVR on increasing self-compassion and reduce self-criticism, thus better improving symptoms of depression.

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